Engineered EMI/RFI Shielding Glass for Data Centers, SCIFs & Federal Facilities

Hotlineglass USA engineers transparent EMI/RFI shielding glazing systems as complete electrical assemblies rather than simply laminating conductive films into glass. Our multi-layer architecture integrates conductive films, engineered busbars, engineered connector systems and controlled lamination processes to support consistent electrical performance, long-term durability and verifiable quality.

Why Engineering Matters

Shielding performance depends on the complete electrical path—including conductive layers, busbars, connectors, grounding strategy and manufacturing quality—not just the sheet resistance of a film.

HLG Multi-Layer Architecture

Depending on project requirements, HLG assemblies may incorporate multiple conductive layers, transparent conductive films or meshes, precision conductive interlayers, redundant electrical pathways, engineered busbars and application-specific connector systems.

Engineered Busbars

Engineered busbars improve current collection, reduce contact resistance, minimize voltage gradients, improve shielding uniformity and enhance long-term reliability.

Engineered Connector Systems

HLG supplies engineered connector systems, including HLG exclusive designs, developed for low electrical resistance, mechanical durability, corrosion resistance, simplified installation and reliable grounding.

100% Electrical Inspection

Unlike systems that rely primarily on process validation, HLG designs support electrical verification of every lite, including continuity, resistance, busbar integrity, connector verification and ground-path validation.

Architectural EMI/RFI Security

- AI data centers
- Federal facilities
- SCIFs
- Projects with TEMPEST-related requirements
- Military command centers
- Aerospace
- Secure laboratories
- Financial institutions
- Telecommunications
- Other mission-critical infrastructure

HLG engineers shielded glazing assemblies for projects with TEMPEST-related requirements, built to the customer’s specification and supported by project-specific testing.

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Engineering Advantage

By treating EMI/RFI glazing as an engineered electrical system, Hotlineglass USA supports consistent electrical performance, reduced installation risk and verifiable quality.

Important Note

Actual shielding effectiveness depends on the complete installed system, including glazing construction, framing, grounding, installation methods and project-specific testing. HLG engineers complete shielding solutions to support specified performance objectives.

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